

chessAI

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1 Introduction

The primary objective of **chessAI** is to define a comprehensive set of binary features $f_1, f_2, \dots, f_n$ suitable for fitting a linear regression model. This model serves a dual purpose: first, it functions as a robust evaluation function for the chess engine; second, the resulting weights are utilized to quantify specific strategic aspects, providing explainable analysis for human players.

2 ChessAI Features

All features are calculated from the perspective of the active player (the side currently to move). This relative framing eliminates the need for explicit “tempo” or “turn” features, thereby enhancing model robustness and mitigating potential color bias. For the remainder of this context, we define **White** as the active player and **Black** as the opponent.

2.1 Material Features

All features are calculated from the perspective of the active player (White).

2.1.1 Pawns

16 binary features. We differentiate pawns based on their starting file.

$$P_f^c = \begin{cases} 1 & \text{if pawn of color } c \text{ from file } f \text{ is on the board,} \\ 0 & \text{otherwise.} \end{cases}$$

2.1.2 Knights

2 binary features per player. Since knights can move to any square color and are interchangeable, we define them by count threshold rather than flank.

- N_1^c : Indicates the presence of at least one Knight.

$$N_1^c = 1 \quad \text{if Knight count for } c \geq 1.$$

- N_2^c : Indicates the presence of a second Knight.

$$N_2^c = 1 \quad \text{if Knight count for } c = 2.$$

2.1.3 Bishops

2 binary features per player. Bishops are distinct by square color.

$$B_{sq}^c = 1 \quad \text{if the } sq\text{-squared Bishop of color } c \text{ is on the board.}$$

where $sq \in \{\text{Light}, \text{Dark}\}$.

2.1.4 Rooks

2 binary features per player. Similar to knights, rooks are interchangeable and defined by existence thresholds.

- R_1^c : Indicates the presence of at least one Rook.

$$R_1^c = 1 \quad \text{if Rook count for } c \geq 1.$$

- R_2^c : Indicates the presence of a second Rook.

$$R_2^c = 1 \quad \text{if Rook count for } c = 2.$$

2.1.5 Queen

1 binary feature per player.

$$Q^c = 1 \quad \text{if Queen of color } c \text{ is on the board.}$$

2.1.6 Bishop Pair

1 binary feature per player.

$$\text{Pair}_B^c = 1 \quad \text{if } B_{\text{Light}}^c = 1 \text{ and } B_{\text{Dark}}^c = 1.$$

2.1.7 Knight Pair

1 binary feature per player. Note: This feature is mathematically identical to N_2^c in this encoding but is retained for explicit weight analysis of the "pair synergy."

$$\text{Pair}_N^c = 1 \quad \text{if } N_2^c = 1.$$

2.1.8 Rook Pair

1 binary feature per player. Similar to the Knight pair.

$$\text{Pair}_R^c = 1 \quad \text{if } R_2^c = 1.$$

2.1.9 No Pawns

1 binary feature per player.

$$\text{NoPawns}^c = 1 \quad \text{if} \quad \sum_{f=a}^h P_f^c = 0.$$

2.1.10 Material Imbalances

These features capture specific non-standard material trade-offs. We define them for the Active Player (c) holding the pieces specified first, against the Opponent ($\bar{c}$) holding the pieces specified second.

- **Queen vs. 3 Minors:** 1 feature per player. Indicates the player has a Queen while the opponent has 3 minor pieces (Knights/Bishops).

$$\text{Imb}_{Qv3M}^c = 1 \quad \text{if} \quad Q^c = 1 \wedge (N^{\bar{c}} + B^{\bar{c}} \geq 3).$$

- **Queen vs. 2 Rooks:** 1 feature per player.

$$\text{Imb}_{Qv2R}^c = 1 \quad \text{if} \quad Q^c = 1 \wedge R^{\bar{c}} = 2.$$

- **Rook vs. 2 Minors:** 1 feature per player. Indicates the player has a Rook while the opponent has 2 minor pieces (often a disadvantage for the Rook holder).

$$\text{Imb}_{Rv2M}^c = 1 \quad \text{if} \quad R^c = 1 \wedge (N^{\bar{c}} + B^{\bar{c}} = 2).$$

- **Bishop vs. Knight:** 1 feature per player. This feature is active if the player possesses a Bishop(s) in an endgame scenario against an opponent with Knight(s), with no other minor pieces complicating the imbalance.

$$\text{Imb}_{BvN}^c = 1 \quad \text{if} \quad B^c \geq 1 \wedge N^{\bar{c}} \geq 1 \wedge N^c = 0 \wedge B^{\bar{c}} = 0.$$

2.2 Piece-Square Table (PST) Features

These features capture spatial control and piece activity. To allow the model to distinguish ownership (e.g., “My Knight on e5” vs. “Opponent Knight on e5”), we generate distinct feature sets for the active player (White) and the opponent (Black).

2.2.1 Pawn Position

Total: 96 binary features (48×2). We exclude ranks 1 and 8 as valid pawn positions.

- **Active Player (White):** 48 features.

$$P_s^{\text{White}} = 1 \quad \text{if a White pawn is on square } s.$$

- **Opponent (Black):** 48 features.

$$P_s^{Black} = 1 \quad \text{if a Black pawn is on square } s.$$

where $s \in \{a2, \dots, h7\}$.

2.2.2 Piece Position

For non-pawn pieces, we define 128 binary features per piece type (64×2), covering every square on the board. Let $s \in \{a1, \dots, h8\}$.

- **Knights:** 128 features.

$$N_s^c = 1 \quad \text{if a Knight of color } c \text{ is on square } s.$$

- **Bishops:** 128 features.

$$B_s^c = 1 \quad \text{if a Bishop of color } c \text{ is on square } s.$$

- **Rooks:** 128 features.

$$R_s^c = 1 \quad \text{if a Rook of color } c \text{ is on square } s.$$

- **Queens:** 128 features.

$$Q_s^c = 1 \quad \text{if a Queen of color } c \text{ is on square } s.$$

- **Kings:** 128 features.

$$K_s^c = 1 \quad \text{if the King of color } c \text{ is on square } s.$$

where $c \in \{\text{White}, \text{Black}\}$.

2.3 Pawn Structure Features

These features capture the static and dynamic properties of the pawn skeleton. As structure is color-dependent (e.g., an isolated pawn is a weakness for the owner), we generate separate feature sets for the Active Player (c) and the Opponent ($\bar{c}$).

2.3.1 File-Based Structural Features

These features identify structural weaknesses or properties on specific files. Total: $8 \times 2 = 16$ features per category.

Let $Files = \{a, b, \dots, h\}$. For a file f , let $Adj(f)$ be the set of adjacent files.

- **Isolated Pawn:** A pawn with no friendly pawns on adjacent files.

$$Iso_f^c = 1 \quad \text{if } \exists r, P_{f,r}^c = 1 \wedge \forall f' \in Adj(f), \sum_r P_{f',r}^c = 0.$$

- **Doubled Pawn:** More than one pawn of the same color on a single file.

$$\text{Double}_f^c = 1 \quad \text{if } \sum_r P_{f,r}^c \geq 2.$$

- **Backward Pawn:** A pawn that is behind its neighbors and cannot safely advance.

$$\text{Back}_f^c = 1 \quad \text{if } P_{f,r}^c = 1 \text{ is the rearmost pawn and support from } \text{Adj}(f).$$

- **Weak Chain Base:** The base of a pawn chain is the rearmost pawn in a connected diagonal chain, making it the structural weak point.

$$\text{WeakBase}_f^c = 1 \quad \text{if pawn on } f \text{ supports another but is not supported itself.}$$

- **Overextended Pawn:** A pawn advanced beyond its support (typically rank 5 or 6) that is vulnerable.

$$\text{Over}_f^c = 1 \quad \text{if } P_{f,r}^c = 1 \text{ where } r \geq 5 \text{ and has no pawn support.}$$

- **Backward on Open File:** A backward pawn on a file where the player has no other pawns and the file is open or semi-open.

$$\text{BackOpen}_f^c = 1 \quad \text{if } \text{Back}_f^c = 1 \text{ and file } f \text{ is semi-open.}$$

- **Pawn Ram:** Total $8 \times 2 = 16$ features. A "Ram" occurs when a pawn is directly blocked by an enemy pawn on the square immediately in front of it, creating a deadlock on that file.

$$\text{Ram}_f^c = 1 \quad \text{if } P_{f,r}^c = 1 \wedge P_{f,r+1}^{\bar{c}} = 1.$$

- **Pawn Lever (Break):** Total $8 \times 2 = 16$ features. A "Lever" is a tension pattern where a pawn attacks an enemy pawn diagonally, creating the possibility to capture and open a line (a "pawn break").

$$\text{Lever}_f^c = 1 \quad \text{if } P_{f,r}^c = 1 \wedge (P_{f-1,r+1}^{\bar{c}} = 1 \vee P_{f+1,r+1}^{\bar{c}} = 1).$$

- **Self-Blocked Pawn:** Total $8 \times 2 = 16$ features. A pawn that is unable to move because the square immediately in front is occupied by a *friendly* piece.

$$\text{SelfBlock}_f^c = 1 \quad \text{if } P_{f,r}^c = 1 \wedge \exists \text{ piece } p^c \text{ on } (f, r+1).$$

2.3.2 Passed Pawn Features

Passed pawns are critical winning factors. We define these features per square (File f , Rank r) for ranks 2–7. Total: $48 \times 2 = 96$ features per category.

- **Passed Pawn:** A pawn with no opposing pawns in front on the same or adjacent files.

$$\text{Pass}_{f,r}^c = 1 \quad \text{if } P_{f,r}^c = 1 \wedge \forall r' > r, f' \in \{f\} \cup \text{Adj}(f) : P_{f',r'}^{\bar{c}} = 0.$$

- **Protected Passer:** A passed pawn that is simultaneously defended by another friendly pawn.

$$\text{ProtPass}_{f,r}^c = 1 \quad \text{if } \text{Pass}_{f,r}^c = 1 \wedge \text{IsDefended}(P_{f,r}^c).$$

- **Unstoppable Passer:** A passed pawn where the enemy King cannot enter the "square of the pawn" to stop promotion.

$$\text{Unstop}_{f,r}^c = 1 \quad \text{if } \text{Pass}_{f,r}^c = 1 \wedge \text{Dist}(P_{f,r}^c, \text{Prom}) < \text{Dist}(K^{\bar{c}}, \text{Prom}).$$

- **Passed Pawn Race Advantage:** 1 feature per player. In a mutual pawn race (both sides have passed pawns), this feature indicates if the active player promotes faster.

$$\text{RaceAdv}^c = 1 \quad \text{if } \text{MovesToPromote}(P^c) < \text{MovesToPromote}(P^{\bar{c}}).$$

- **Tarrasch Rule Compliance:** 1 feature per player. Captures the strategic principle that "Rooks belong behind passed pawns."

$$\text{Tarrasch}^c = 1 \quad \text{if } R^c \text{ is on the same file and behind } P^c \text{ (supporting) OR } P^{\bar{c}} \text{ (stopping)}.$$

- **King in Pawn Square:** 1 feature per player. Indicates if the defending King is inside the "Square of the Pawn" of the opponent, theoretically able to stop it (if no other pieces interfere).

$$\text{InSquare}^{\bar{c}} = 1 \quad \text{if } K^{\bar{c}} \in \text{Square}(P^c).$$

- **King Support (Near Passer):** 1 feature per player. Indicates if the friendly King is close enough to support the advance of their passed pawn.

$$\text{KingSupport}^c = 1 \quad \text{if } \text{Dist}(K^c, P^c) \leq 2.$$

- **King Blocking Proximity:** 1 feature per player. A binary measure of the King's ability to intercept an enemy passed pawn.

$$\text{CloseBlocker}^{\bar{c}} = 1 \quad \text{if } \text{Dist}(K^{\bar{c}}, \text{Path}(P^c)) \leq 3.$$

- **Connected Passers:** 48 features per player (per square). Connected passed pawns are exceptionally strong as they protect each other while advancing.

$$\text{ConnPass}_{f,r}^c = 1 \quad \text{if } \text{Pass}_{f,r}^c = 1 \wedge \exists \text{ Passed Pawn of } c \text{ on file } f \pm 1.$$

- **Candidate Passer:** Total $8 \times 2 = 16$ features. A "Candidate" is a pawn that is not yet passed but has the potential to become one because it is part of a healthy majority (no opposing pawns on the file, or the player has a force advantage on that wing).

$$\text{Cand}_f^c = 1 \quad \text{if } P_f^c \text{ is not passed} \wedge \text{CanForcePass}(P_f^c).$$

2.3.3 Connectivity Features

These features capture horizontal and diagonal pawn coordination. Total: $7 \times 2 = 14$ features per category (defined on adjacent file pairs $f, f+1$).

- **Connected Pawns:** Pawns exist on adjacent files f and $f+1$.

$$\text{Conn}_f^c = 1 \quad \text{if } (\sum P_{f,r}^c \geq 1) \wedge (\sum P_{f+1,r}^c \geq 1).$$

- **Phalanx:** Two pawns on adjacent files on the *same rank* r , controlling the squares ahead.

$$\text{Phalanx}_f^c = 1 \quad \text{if } \exists r : P_{f,r}^c = 1 \wedge P_{f+1,r}^c = 1.$$

2.3.4 Complex Structural Features

These features identify specific pawn formations.

- **Chain Set (Elementary Links):** Total $70 \times 2 = 140$ binary features. A "chain set" represents the fundamental building block of a pawn chain: a single protection relationship between two pawns. We generate a distinct feature for every possible specific link on the board.

Let $\text{Link}_{f,r}^{dir,c}$ be a binary feature where $dir \in \{\text{Left}, \text{Right}\}$.

- **Right Link** ($f \rightarrow f+1$):

$$\text{Link}_{f,r}^{\text{Right},c} = 1 \quad \text{if } P_{f,r}^c = 1 \wedge P_{f+1,r+1}^c = 1.$$

Valid for $f \in \{a, \dots, g\}$ and $r \in \{2, \dots, 6\}$.

- **Left Link** ($f \rightarrow f-1$):

$$\text{Link}_{f,r}^{\text{Left},c} = 1 \quad \text{if } P_{f,r}^c = 1 \wedge P_{f-1,r+1}^c = 1.$$

Valid for $f \in \{b, \dots, h\}$ and $r \in \{2, \dots, 6\}$.

- **Pawn Islands Shapes:** Total $36 \times 2 = 72$ binary features. An island is defined as a contiguous set of files containing pawns, separated by files with no pawns. We generate a feature for every possible "island shape" (a specific range of files).

Let S be a contiguous segment of files (e.g., $\{a, b\}$ or $\{c, d, e\}$).

$$\text{Island}_S^c = 1 \quad \text{if } (\forall f \in S, \sum_r P_{f,r}^c \geq 1) \wedge (\forall f \notin S \text{ adjacent to } S, \sum_r P_{f,r}^c = 0).$$

2.3.5 Global Counts

Bucketized features to capture the magnitude of structural elements. Total: $3 \times 2 = 6$ features per category.

- **Pawn Chain Count:** One-hot encoding for the total number of elementary links (Chain Sets) on the board.

$$\text{ChainCount}_k^c = 1 \quad \text{if } (\sum \text{Link}^c) = k, \quad \text{for } k \in \{1, 2, 3+\}.$$

- **Pawn Island Count:** One-hot encoding for the number of distinct islands.

$$\text{IslandCount}_k^c = 1 \quad \text{for } k \in \{1, 2, 3+\}.$$

2.3.6 Wing Distribution and Majorities

These features capture the global distribution of material across the board, which dictates long-term strategy (e.g., creating a passed pawn where you have a majority).

- **Wing Presence:** 2 features per player.
 - $\text{BothWings}^c = 1$: If player c has pawns on both the Queenside (files a–c) and Kingside (files f–h).
 - $\text{OneWing}^c = 1$: If player c has pawns on only one flank (Kingside OR Queenside) and none on the other.
- **Pawn Majorities:** 3 features per player. A majority exists if the player has more pawns in a specific sector than the opponent.

$$\text{Maj}_{Kingside}^c = 1: \text{ If } \sum_{f=f}^h P_f^c > \sum_{f=f}^h P_f^{\bar{c}}.$$

$$\text{Maj}_{Queenside}^c = 1: \text{ If } \sum_{f=a}^c P_f^c > \sum_{f=a}^c P_f^{\bar{c}}.$$

$$\text{Maj}_{Central}^c = 1: \text{ If } \sum_{f=d}^e P_f^c > \sum_{f=d}^e P_f^{\bar{c}}.$$

- **Crippled Majority:** 3 features per player.

$$\text{CrippledMaj}_{sector}^c = 1 \quad \text{if } \text{Maj}_{sector}^c = 1 \wedge \text{structure in } sector \text{ contains a doubled or backward pawn.}$$

2.3.7 Dynamic Pawn Features

- **En Passant Vulnerability/Opportunity:** 1 feature per player.

$$\text{EnPassant}^c = 1 \quad \text{if } c \text{ has a valid en passant capture available.}$$

2.4 King Safety Features

King safety is critical and highly context-dependent. We define these features relative to the position of the King (K_{pos}^c). Total features are calculated for both the Active Player (c) and Opponent ($\bar{c}$).

2.4.1 Pawn Shield

These features evaluate the protective wall in front of the King. They are only active if the King is on the 1st Rank (relative to color). Let $file(K)$ be the King's file. We examine the three relative files: $f \in \{file(K)-1, file(K), file(K)+1\}$.

- **Rank 2 Shield:** 3 features per player.

$$\text{Shield}_{2,dir}^c = 1 \quad \text{if } P_{f,rank=2}^c = 1 \text{ and King is on rank 1.}$$

where $dir \in \{\text{Left}, \text{Center}, \text{Right}\}$.

- **Rank 3 Shield:** 3 features per player.

$$\text{Shield}_{3,dir}^c = 1 \quad \text{if } P_{f,rank=3}^c = 1 \text{ and King is on rank 1.}$$

- **Shield Under Attack:** 3 features per player. Indicates if any of the specific shield pawns are currently threatened by enemy pieces.

$$\text{ShieldAttack}_{dir}^c = 1 \quad \text{if shield pawn at } dir \text{ is under attack.}$$

- **Shield Ram:** 3 features per player. A "Ram" occurs when a shield pawn is blocked directly by an enemy pawn immediately in front of it (e.g., White pawn on g2, Black pawn on g3). This often indicates an incoming pawn storm or tension.

$$\text{ShieldRam}_{dir}^c = 1 \quad \text{if shield pawn at } dir \text{ is blocked by an enemy pawn.}$$

2.4.2 King Position Environment

- **King Shelter:** 8 features per player. Indicates if the 8 squares surrounding the King are occupied by friendly pieces (providing shelter from checks).

$$\text{Shelter}_{dir}^c = 1 \quad \text{if neighbor square } dir \text{ contains a friendly piece.}$$

where $dir \in \{\text{Top}, \text{TopRight}, \dots, \text{TopLeft}\}$.

- **King Virtual Mobility:** 28 features per player. A heuristic measure of the King's openness. We treat the King as a Queen and calculate the number of pseudo-legal moves (m) it could make on an empty board (or considering current blockers).

$$\text{VirtMob}_k^c = 1 \quad \text{if virtual moves } m = k, \quad \text{for } k \in \{0, \dots, 27\}.$$

- **King Back Rank:** 1 feature per player.

$$\text{BackRank}^c = 1 \quad \text{if King is on rank 1.}$$

- **King on Open File:** 1 feature per player.

$$\text{OpenKing}^c = 1 \quad \text{if no pawns (friendly or enemy) exist on the King's file.}$$

- **King on Semi-Open File:** 1 feature per player.

$$\text{SemiOpenKing}^c = 1 \quad \text{if no friendly pawns exist on the King's file.}$$

2.4.3 Structural Weaknesses

- **Fianchetto Holes (Missing Bishop):** 2 features per player. This feature identifies the dangerous pattern where the Knight pawn is advanced (fianchetto structure), creating weak squares ("holes"), but the Bishop is not present to defend them.

$$\text{Holes}_{flank}^c = 1 \quad \text{if } P_{N-file}^c \text{ is advanced} \wedge \text{Bishop}_{flank}^c \text{ is absent.}$$

where $flank \in \{\text{Kingside}, \text{Queenside}\}$.

2.4.4 King Zone Pressure and Defense

These features quantify the intensity of the attack and the strength of the defensive shell. We define the "King Zone" as the set of squares adjacent to the King plus the squares immediately in front of them.

- **King Zone Attackers Count:** 3 features per player. We count the number of distinct enemy pieces attacking at least one square in the King Zone.

$$\text{ZoneAttackers}_k^c = 1 \quad \text{if count of attacking pieces} = k, \quad \text{for } k \in \{1, 2, 3+\}.$$

- **King Zone Defenders Count:** 3 features per player.

$$\text{ZoneDefenders}_k^c = 1 \quad \text{if count of friendly pieces defending squares in the King Zone} = k, \quad \text{for } k \in \{1, 2, 3+\}.$$

2.5 Piece Movement Features

These features capture the activity level of the pieces. To reflect true piece effectiveness, we calculate **Safe Mobility**: the count of pseudo-legal moves to squares that are *not* currently attacked by the opponent.

Let $SafeMoves(p)$ be the count of pseudo-legal moves for piece p to squares s where $IsAttacked(s, \bar{c}) = \text{False}$.

2.5.1 Mobility Buckets

For each piece type, we generate binary features corresponding to exact safe move counts.

- **Knight Safe Mobility:** 9 features per player (Range 0–8).

$$\text{Mob}_{N,k}^c = 1 \quad \text{if } \exists n \in \text{Knights}^c \text{ s.t. } SafeMoves(n) = k.$$

- **Bishop Safe Mobility:** 14 features per player (Range 0–13).

$$\text{Mob}_{B,k}^c = 1 \quad \text{if } \exists b \in \text{Bishops}^c \text{ s.t. } SafeMoves(b) = k.$$

- **Rook Safe Mobility:** 15 features per player (Range 0–14).

$$\text{Mob}_{R,k}^c = 1 \quad \text{if } \exists r \in \text{Rooks}^c \text{ s.t. } SafeMoves(r) = k.$$

- **Queen Safe Mobility:** 28 features per player (Range 0–27).

$$\text{Mob}_{Q,k}^c = 1 \quad \text{if } SafeMoves(Q^c) = k.$$

- **King Safe Mobility:** 9 features per player (Range 0–8). Note: For the King, "Safe Mobility" is equivalent to legal moves (squares not in check).

$$\text{Mob}_{K,k}^c = 1 \quad \text{if } SafeMoves(K^c) = k.$$

2.5.2 Castling Features

Features describing the castling status and potential.

- **Castling Rights:** 2 features per player.

$$\text{CanCastle}_{side}^c = 1 \quad \text{if rights to castle } side \text{ exist.}$$

where $side \in \{\text{Short}, \text{Long}\}$.

- **Castling Status:** 2 features per player.

$$\text{IsCastled}_{side}^c = 1 \quad \text{if player } c \text{ has castled } side.$$

- **Lost Castling Rights:** 1 feature per player.

$$\text{NoCastle}^c = 1 \quad \text{if player has lost all castling rights and has not castled.}$$

2.5.3 Piece Restrictions and Strategic Placement

These features identify specific strategic patterns where pieces are either optimally placed (good) or severely restricted (bad).

- **Trapped Bishop (Corner):** 2 features per player. Identifies if a Bishop is trapped on a corner square (e.g., h1/a8) by its own pawns.

$$\text{Trap}_{B,sq}^c = 1 \quad \text{if } B_{sq}^c = 1 \text{ (corner)} \wedge \text{Mobility}(B) < 2.$$

- **Trapped Rook (Boxed):** 2 features per player. Identifies a Rook that is completely "boxed in" by its own pieces (often the King) with zero mobility.

$$\text{Trap}_{R,flank}^c = 1 \quad \text{if } R_{flank}^c = 1 \wedge \text{Mobility}(R) = 0.$$

- **Trapped Knight (Rim/Corner):** 1 feature per player.

$$\text{Trap}_{N,sq}^c = 1 \quad \text{if } N^c \text{ is on an edge square} \wedge \text{SafeMoves}(N^c) = 0.$$

- **Trapped Queen:** 1 feature per player.

$$\text{Trap}_Q^c = 1 \quad \text{if } \text{SafeMoves}(Q^c) = 0 \wedge Q^c \text{ is deep in enemy territory (ranks 6–8 relative).}$$

- **Bad Bishop (Blocked Center):** 2 features per player. A Bishop located centrally but blocked by its own central pawns.

$$\text{Bad}_{B,sq}^c = 1 \quad \text{if } B^c \text{ is central} \wedge \text{blocked by own pawns.}$$

- **Bishop Blocked by Own Pawns:** 1 feature per player. Often called a "Bad Bishop" structurally.

$$\text{Block}_{B,Own}^c = 1 \quad \text{if } B^c \text{ exists} \wedge \text{resides on same color as} \geq 3 \text{ friendly central pawns.}$$

- **Bishop Fixing Enemy Pawns:** 1 feature per player. A good Bishop that attacks the base of an enemy pawn chain, fixing the pawns to that color (making them vulnerable).

$$\text{Fixing}_B^c = 1 \quad \text{if } B^c \text{ attacks enemy chain base on matching color.}$$

- **Fianchetto:** 2 features per player. Indicates if a Bishop is developed to the long diagonal inside a pawn notch (e.g., Bg2 with pawn on g3).

$$\text{Fianchetto}_{flank}^c = 1 \quad \text{if } B^c \text{ is on } b2/g2 \text{ (or } b7/g7) \wedge P_{b/g}^c \text{ is advanced.}$$

- **Cramped Position:** 1 feature per player. A global indicator of restricted space.

$$\text{Cramped}^c = 1 \quad \text{if Total Mobility for } c < \text{Threshold.}$$

- **Bishop Dominant Color:** 2 features per player. Indicates if a Bishop controls a color complex that is weak for the opponent (e.g., opponent is missing that Bishop).

$$\text{Dom}_{B,color}^c = 1 \quad \text{if } B_{color}^c = 1 \wedge B_{color}^{\bar{c}} = 0.$$

- **Bishop Passive Color:** 2 features per player. Indicates if a Bishop is rendered passive because its own pawns fixed on its color obstruct it.

$$\text{Passive}_{B,color}^c = 1 \quad \text{if } B_{color}^c = 1 \wedge \text{Majority of own pawns are on } color.$$

2.6 Pins and Skewers

These features identify tactical immobilization. A "pin" occurs when an enemy piece cannot move without exposing a higher-value target. A "skewer" is the inverse, attacking a high-value target to expose a piece behind it. All features are defined for the active player c (the pinner/attacker).

2.6.1 Absolute Pins (to King)

An absolute pin exists when a piece is pinned against its King. We distinguish these by the geometry of the pin and the attacking piece type.

- **Orthogonal Absolute Pin:** 2 features per player. Defined for files and ranks.
 - $\text{Pin}_{Abs,Rook}^c = 1$: If a Rook of color c pins an enemy piece to the King.
 - $\text{Pin}_{Abs,Queen}^c = 1$: If a Queen of color c pins an enemy piece to the King orthogonally.
- **Diagonal Absolute Pin:** 2 features per player. Defined for diagonals.
 - $\text{Pin}_{Abs,Bishop}^c = 1$: If a Bishop of color c pins an enemy piece to the King.
 - $\text{Pin}_{Abs,Queen}^c = 1$: If a Queen of color c pins an enemy piece to the King diagonally.

2.6.2 Relative Pins (to Major Pieces)

A relative pin exists when a piece is pinned against a high-value piece other than the King (specifically Queen or Rook). 2 features per player.

- **Pin to Queen:**

$$\text{Pin}_{Rel,Q}^c = 1 \quad \text{if a piece of color } c \text{ pins an enemy piece to the enemy Queen.}$$

- **Pin to Rook:**

$$\text{Pin}_{Rel,R}^c = 1 \quad \text{if a piece of color } c \text{ pins an enemy piece to an enemy Rook.}$$

2.6.3 Soft Pins (Tactical Tension)

A "soft pin" occurs when a piece is pinned to an undefended piece of equal or lower value.

- **Soft Pin:** 1 feature per player.

$$Pin_{Soft}^c = 1 \quad \text{if moving a piece of } \bar{c} \text{ exposes an attack on a loose/hanging piece.}$$

2.6.4 Advanced Tactics

- **Cross Pin:** 1 feature per player. This occurs when a single enemy piece is pinned from two different directions simultaneously (e.g., pinned vertically by a Rook and diagonally by a Bishop).

$$Pin_{Cross}^c = 1 \quad \text{if an enemy piece is subject to } \geq 2 \text{ distinct pins by } c.$$

- **King Skewer:** 1 feature per player. A skewer against the King occurs when the King is attacked (checked) and forced to move, exposing a piece behind it on the line of attack.

$$Skewer_K^c = 1 \quad \text{if } c \text{ checks } K^{\bar{c}} \text{ and an enemy piece exists behind } K^{\bar{c}} \text{ on the attack ray.}$$

2.7 Batteries

A battery is formed by two or more pieces of the same color lined up on a single rank, file, or diagonal, multiplying the force of an attack. All features are binary and calculated for the active player c .

2.7.1 Vertical Batteries (Files)

Batteries formed on files (columns), typically targeting the enemy back rank or pawns.

- **Rook-Rook Vertical Battery:** 1 feature per player.

$$Bat_{Ver,RR}^c = 1 \quad \text{if } \exists \text{ file } f \text{ containing 2 Rooks of color } c.$$

- **Queen-Rook Vertical Battery:** 1 feature per player.

$$Bat_{Ver,QR}^c = 1 \quad \text{if } \exists \text{ file } f \text{ containing a Queen and a Rook of color } c.$$

2.7.2 Horizontal Batteries (Ranks)

Batteries formed on ranks, often useful for defense or 7th rank invasions.

- **Rook-Rook Horizontal Battery:** 1 feature per player.

$$Bat_{Hor,RR}^c = 1 \quad \text{if } \exists \text{ rank } r \text{ containing 2 Rooks of color } c.$$

2.7.3 Diagonal Batteries

Batteries formed on diagonals, primarily targeting the King or long-range weaknesses.

- **Queen-Bishop Battery:** 1 feature per player.

$$\text{Bat}_{\text{Diag},QB}^c = 1 \quad \text{if } \exists \text{ diagonal containing a Queen and a Bishop of color } c.$$

2.7.4 Special Formations

- **Alekhine’s Gun:** 1 feature per player. A rare and powerful triple-piece battery on a single file, consisting of two Rooks and a Queen (typically arranged R-R-Q from front to back to maximize penetration power).

$$\text{Alekhine}^c = 1 \quad \text{if } \exists \text{ file } f \text{ containing 2 Rooks and 1 Queen of color } c.$$

2.8 Knight Geometry

These features evaluate the spatial relationship between enemy high-value pieces to identify potential vulnerabilities to Knight forks. We define "Fork Geometry" as a specific distance between two pieces that allows a Knight to theoretically attack both simultaneously (regardless of whether a Knight is currently in position to do so).

2.8.1 Geometric Vulnerabilities (Fork Distance)

These features are 1 if the two specified enemy pieces are positioned on squares compatible with a standard Knight fork (e.g., separation allows for a simultaneous attack). All features are calculated for the Active Player (c) scanning the Opponent’s pieces ($\bar{c}$).

- **King-Queen Fork Geometry:** 1 feature.

$$\text{Geom}_{K,Q}^c = 1 \quad \text{if } K^{\bar{c}} \text{ and } Q^{\bar{c}} \text{ are at a forkable distance.}$$

- **King-Rook Fork Geometry:** 1 feature.

$$\text{Geom}_{K,R}^c = 1 \quad \text{if } K^{\bar{c}} \text{ and any } R^{\bar{c}} \text{ are at a forkable distance.}$$

- **Queen-Rook Fork Geometry:** 1 feature.

$$\text{Geom}_{Q,R}^c = 1 \quad \text{if } Q^{\bar{c}} \text{ and any } R^{\bar{c}} \text{ are at a forkable distance.}$$

- **King-Bishop Fork Geometry:** 1 feature.

$$\text{Geom}_{K,B}^c = 1 \quad \text{if } K^{\bar{c}} \text{ and any } B^{\bar{c}} \text{ are at a forkable distance.}$$

- **Rook-Rook Fork Geometry:** 1 feature.

$$\text{Geom}_{R,R}^c = 1 \quad \text{if two } R^{\bar{c}} \text{ pieces are at a forkable distance.}$$

2.8.2 Tactical Execution

- **Knight Fork with Check:** 1 feature. An immediate tactical pattern where a Knight simultaneously attacks the King (Check) and another piece.

$\text{ForkCheck}^c = 1$ if a Knight of color c checks $K^{\bar{c}}$ and attacks another piece $p^{\bar{c}}$.

2.9 Direct Threats and Vulnerabilities

These features quantify immediate tactical dangers and opportunities. We define them from the perspective of the Active Player (c) creating threats against the Opponent ($\bar{c}$), or identifying weaknesses in the Opponent's position.

2.9.1 Hanging and Loose Pieces

A "Hanging" piece is one that is attacked and undefended (or attacked by a lower-value piece). A "Loose" piece is one that is completely undefended, regardless of whether it is currently attacked.

- **Hanging Pieces:** 5 binary features (one per piece type).

$\text{Hang}_{type}^{\bar{c}} = 1$ if a piece of $type \in \{P, N, B, R, Q\}$ belonging to $\bar{c}$ is hanging.

- **Loose Piece Count:** 2 binary features (buckets). These features count the total number of undefended pieces in the opponent's camp.
 - $\text{Loose}_1^{\bar{c}} = 1$: If exactly 1 enemy piece is loose.
 - $\text{Loose}_{2+}^{\bar{c}} = 1$: If 2 or more enemy pieces are loose.

2.9.2 Discovery and X-Ray

Features involving "batteries" where the front piece moves to unmask an attack.

- **Discovery Potential (King):** 1 feature.

$\text{Disc}_K^c = 1$ if moving a piece p^c unmask a sliding attack on $K^{\bar{c}}$.

- **Discovery Potential (Queen):** 1 feature.

$\text{Disc}_Q^c = 1$ if moving a piece p^c unmask a sliding attack on $Q^{\bar{c}}$.

- **X-Ray Attack (King):** 1 feature. An X-Ray occurs when a piece attacks a square occupied by an enemy piece, and the King is behind that piece on the same line.

$\text{XRay}_K^c = 1$ if c attacks a piece blocking the line to $K^{\bar{c}}$.

2.9.3 Specific Tactical Threats

- **Pawn Fork:** 1 feature.

$\text{Fork}_P^c = 1$ if a Pawn of c attacks two enemy pieces simultaneously.

- **Minor Attacked by Pawn:** 1 feature. A high-value trade threat.

$\text{Threat}_{m,P}^c = 1$ if a Minor piece (N/B) of $\bar{c}$ is attacked by a Pawn of c .

- **Rook Attacked by Minor:** 1 feature.

$\text{Threat}_{R,m}^c = 1$ if a Rook of $\bar{c}$ is attacked by a Minor piece of c .

- **Double Attack (Sliding Piece):** 1 feature per player.

$\text{DoubleAttack}_p^c = 1$ if a sliding piece p^c simultaneously attacks two undefended enemy pieces (or pieces

- **Check:** 1 feature.

$\text{IsCheck}^c = 1$ if $K^{\bar{c}}$ is currently under attack.

2.9.4 Overloaded Defenders

A defender is "overloaded" if it is tasked with defending multiple attacked pieces or key squares simultaneously. Total: 7 binary features (1 Queen, 2 Rooks, 2 Bishops, 2 Knights).

$\text{Overload}_p^{\bar{c}} = 1$ if piece p of $\bar{c}$ defends ≥ 2 distinct threatened targets.

Defined for each specific non-pawn piece:

- $Q^{\bar{c}}$ (1 feature)
- $R_{\text{Queenside}}^{\bar{c}}, R_{\text{Kingside}}^{\bar{c}}$ (2 features)
- $B_{\text{Light}}^{\bar{c}}, B_{\text{Dark}}^{\bar{c}}$ (2 features)
- $N_1^{\bar{c}}, N_2^{\bar{c}}$ (2 features)

2.10 Attacking Patterns

These features identify high-level strategic motifs and recognized mating patterns. Features are calculated for the Active Player (c) attacking the Opponent ($\bar{c}$).

2.10.1 Mating Patterns

- **Greek Gift Potential:** 1 feature per player. Identifies the conditions for the classic Bishop sacrifice on h7 (or h2).

$$\text{GreekGift}^c = 1 \quad \text{if } B^c \text{ attacks } h7/h2 \text{ (relative)} \wedge N^c \text{ can jump to } g5/g4.$$

- **Safe Check Available:** 1 feature per player. Indicates if the player has a move that delivers check without immediately losing the checking piece (a "spite check" is not a safe check).

$$\text{SafeCheck}^c = 1 \quad \text{if } \exists \text{ move } m \text{ s.t. } \text{IsCheck}(m) \wedge \neg \text{IsHanging}(m).$$

2.10.2 Pawn Storms

Features indicating an aggressive advance of pawns on a specific flank, typically intended to open lines against the enemy King.

- **Kingside Pawn Storm:** 1 feature per player.

$$\text{Storm}_{Kingside}^c = 1 \quad \text{if multiple pawns on files f-h are advanced (ranks 4-6).}$$

- **Queenside Pawn Storm:** 1 feature per player.

$$\text{Storm}_{Queenside}^c = 1 \quad \text{if multiple pawns on files a-c are advanced (ranks 4-6).}$$

2.10.3 Major Piece Aggression

- **Queen Infiltration (7th Rank):** 1 feature per player. The 7th rank (or 2nd for Black) is often the weakest point in the enemy structure.

$$Q_{7th}^c = 1 \quad \text{if } Q^c \text{ is on rank 7 (if White) or rank 2 (if Black).}$$

2.11 Blockades

A blockade occurs when a piece occupies the square directly in front of an enemy passed pawn, physically preventing its advance. This is a critical defensive concept. Features are defined for the Active Player (c) blocking the Opponent's ($\bar{c}$) passed pawns.

2.11.1 Piece Blockades

Binary features indicating if a specific piece type is performing a blockade.

- **Knight Blockade:** 1 feature. Knights are considered the ideal blockaders as they remain active while blocking.

$$\text{Block}_N^c = 1 \quad \text{if a Knight of } c \text{ is on square } (f, r+1) \text{ blocking } P_{f,r}^{\bar{c}}.$$

- **Bishop Blockade:** 1 feature.

$\text{Block}_B^c = 1$ if a Bishop of c is blocking an enemy passed pawn.

- **Rook Blockade:** 1 feature. Generally considered poor strategy (Rooks should be active), but captured as a feature for evaluation.

$\text{Block}_R^c = 1$ if a Rook of c is blocking an enemy passed pawn.

- **King Blockade:** 1 feature. Essential in endgames.

$\text{Block}_K^c = 1$ if the King of c is blocking an enemy passed pawn.

2.12 Outposts and Weaknesses

These features evaluate the control of key squares. An "Outpost" is typically defined as a piece positioned on an advanced rank (3-6) on an open or semi-open file, which cannot be easily driven away by enemy pawns (a "Hole").

Let $\mathcal{S}_{out}$ be the set of valid outpost squares (28 squares, typically ranks 3-6 excluding edges where applicable).

2.12.1 Piece Outposts

We distinguish between "Protected Outposts" (supported by a friendly pawn) and "Unprotected Outposts" (occupying a hole but relying on piece support). Total: $28 \times 2 = 56$ features per piece type, per player.

- **Knight Outpost:** For each square $s \in \mathcal{S}_{out}$:

- **Protected:** $N_{s,prot}^c = 1$ if a Knight is on s AND supported by a friendly pawn.
- **Unprotected:** $N_{s,unprot}^c = 1$ if a Knight is on s AND NOT supported by a friendly pawn.

- **Bishop Outpost:** For each square $s \in \mathcal{S}_{out}$:

- **Protected:** $B_{s,prot}^c = 1$ if a Bishop is on s AND supported by a friendly pawn.
- **Unprotected:** $B_{s,unprot}^c = 1$ if a Bishop is on s AND NOT supported by a friendly pawn.

2.12.2 Structural Holes

- **Holes:** 28 features per player. A "Hole" is a square in the enemy camp that can no longer be defended by enemy pawns (e.g., because pawns have advanced past it or been captured).

$\text{Hole}_s^{\bar{c}} = 1$ if square s cannot be attacked by any pawn of $\bar{c}$.

where $s \in \mathcal{S}_{out}$.

2.12.3 Color Complex Weaknesses

Features indicating a systematic weakness on squares of a specific color, typically caused by missing the corresponding Bishop or fixing pawns on the opposite color.

- **Weak Dark Complex:** 1 feature per player.

$$\text{Weak}_{\text{Dark}}^{\bar{c}} = 1 \quad \text{if } \bar{c} \text{ has no Dark-square Bishop} \wedge \text{Majority of pawns on Light squares.}$$

- **Weak Light Complex:** 1 feature per player.

$$\text{Weak}_{\text{Light}}^{\bar{c}} = 1 \quad \text{if } \bar{c} \text{ has no Light-square Bishop} \wedge \text{Majority of pawns on Dark squares.}$$

2.13 Development

These features measure the speed at which players bring their pieces into the game. Features are calculated for the Active Player (c).

2.13.1 Development Status

- **Development Lag:** 1 feature per player. Indicates if the player is significantly behind in development compared to the opponent. We define development as the number of non-pawn pieces moved from their starting squares.

$$\text{Lag}^c = 1 \quad \text{if } \text{DevCount}(c) < \text{DevCount}(\bar{c}) - 1.$$

where $\text{DevCount}(p)$ is the count of minor/major pieces for player p that are not on their initial squares.

2.14 Files, Ranks, and Center Control

These features evaluate the control of the board's "highways" (lines) and critical territory. Features are calculated for the Active Player (c).

2.14.1 File Control

- **Open File:** 8 binary features (Board Property). A file is "Open" if it contains no pawns of either color.

$$\text{FileOpen}_f = 1 \quad \text{if } \sum_r (P_{f,r}^{\text{White}} + P_{f,r}^{\text{Black}}) = 0.$$

- **Semi-Open File:** 8 binary features per player. A file is "Semi-Open" for player c if it contains no friendly pawns, but may contain enemy pawns.

$$\text{FileSemi}_f^c = 1 \quad \text{if } \sum_r P_{f,r}^c = 0.$$

- **Rook on Open File:** 8 binary features per player. Indicates if a Rook is optimally placed on a specific open file.

$$\text{RookOpen}_f^c = 1 \quad \text{if } R^c \text{ is on file } f \wedge \text{FileOpen}_f = 1.$$

2.14.2 Rank Invasion and Connectivity

- **Rook on 7th Rank:** 1 feature per player. The "Pig on the 7th" is a powerful attacking motif. (Rank 7 for White, Rank 2 for Black).

$$R_{7th}^c = 1 \quad \text{if } \exists R^c \text{ on relative rank 7.}$$

- **Rook on 8th Rank:** 1 feature per player. Often indicates back-rank mating threats or defense.

$$R_{8th}^c = 1 \quad \text{if } \exists R^c \text{ on relative rank 8.}$$

- **Connected Rooks:** 1 feature per player. Rooks are connected if they are on the same rank or file with no obstructions between them.

$$\text{ConnRooks}^c = 1 \quad \text{if } R_1^c \text{ protects } R_2^c.$$

2.14.3 Center Control

- **Control Center:** 1 feature per player. Indicates if the player controls (attacks) the "Small Center" squares: e4, d4, e5, d5.

$$\text{Center}^c = 1 \quad \text{if } c \text{ attacks } \geq 2 \text{ central squares.}$$

- **Control Extended Center:** 1 feature per player. Indicates control over the larger central ring (c3-f3 to c6-f6).

$$\text{ExtCenter}^c = 1 \quad \text{if } c \text{ attacks majority of extended center squares.}$$

- **King Centrality:** 1 feature per player. A positive feature in the endgame, but often negative in the middlegame.

$$\text{KingCenter}^c = 1 \quad \text{if } K^c \text{ is on e4, d4, e5, or d5.}$$

2.14.4 Space and Maneuverability

- **Space Advantage (Control):** 1 feature per player. A global measure of board dominance.

$$\text{SpaceAdv}^c = 1 \quad \text{if count of squares attacked by } c > 32.$$

- **Sector Space Advantage:** 2 features per player.

$$\text{SpaceAdv}_{Kingside}^c = 1 \quad \text{if squares controlled by } c \text{ on files f-h} > \text{squares controlled by } \bar{c}.$$

$$\text{SpaceAdv}_{Queenside}^c = 1 \quad \text{if squares controlled by } c \text{ on files a-c} > \text{squares controlled by } \bar{c}.$$

- **Long Diagonal Control:** 2 features per player.

$$\text{DiagControl}_{color}^c = 1 \quad \text{if } c \text{ controls more squares on the } color \text{ long diagonal than } \bar{c}.$$

where $color \in \{\text{Light}, \text{Dark}\}$.

- **More Space (Maneuver):** 1 feature per player. Defined by the area available behind the pawn structure for piece movement.

$$\text{Maneuver}^c = 1 \quad \text{if available safe squares behind pawns} > \text{Threshold.}$$

2.15 Interaction Features

To capture non-linear synergies (where the whole is greater than the sum of its parts) within a linear framework, we define interaction features as the product of two binary features.

2.15.1 Material Synergies

Certain piece combinations work exceptionally well together (positive synergy) or poorly together (redundancy).

- **Queen and Knight Attack Force:** 1 feature per player. Captures the known tactical synergy between Queen and Knight (which can fork and attack every color complex).

$$\text{Syn}_{QN}^c = Q^c \times N_1^c$$

- **Rook and Bishop Defense:** 1 feature per player. Rooks and Bishops often coordinate better in defense and open endings than Rooks and Knights.

$$\text{Syn}_{RB}^c = R_1^c \times B_{\text{Light}}^c \text{ (or } B_{\text{Dark}}^c)$$

- **Double Rook Battery:** 1 feature per player. Captures the exponential strength of connected Rooks (already defined as *ConnRooks*, but here explicit as a material multiplier).

$$\text{Syn}_{RR}^c = R_1^c \times R_2^c$$

2.15.2 King Safety Coordination (Attack Multipliers)

To approximate Stockfish's "Attack Units" logic linearly, we flag when high-value pieces **simultaneously** attack the King Zone. Let $\text{Attacks}(p, \text{Zone})$ be true if piece p attacks the enemy King Zone.

- **Queen + Rook Attack:** 1 feature per player.

$$\text{Danger}_{QR}^c = (Q^c \text{ attacks Zone}) \times (\exists R^c \text{ attacking Zone})$$

- **Queen + Knight Attack:** 1 feature per player.

$$\text{Danger}_{QN}^c = (Q^c \text{ attacks Zone}) \times (\exists N^c \text{ attacking Zone})$$

- **Queen + Bishop Attack:** 1 feature per player.

$$\text{Danger}_{QB}^c = (Q^c \text{ attacks Zone}) \times (\exists B^c \text{ attacking Zone})$$

- **Swarm Attack:** 1 feature per player. Active if 3 distinct piece types are attacking the zone simultaneously.

$$\text{Danger}_{\text{Swarm}}^c = 1 \quad \text{if} \quad \sum_{\text{type} \in \{Q, R, B, N\}} I(\text{type attacks Zone}) \geq 3$$

2.15.3 Endgame Scaling Factors

These features allow the model to reduce the evaluation score (predict a draw) even if the player has a material advantage. We create interaction terms between **Total Material** and **Drawish Patterns**.

Let Mat^c be the scalar material score of player c (e.g., $9Q + 5R + 3B + 3N + 1P$).

- **Opposite Color Bishop Dampener:** 1 continuous feature per player. If the game is in the OCB phase, this feature effectively creates a negative weight proportional to the material advantage, pulling the evaluation closer to zero.

$$Scale_{OCB}^c = Mat^c \times Phase_{OCB}$$

- **Fortress Detection (Rook vs Bishop):** 1 feature per player. Identifies the "Wrong Color Bishop" fortress pattern where a Rook pawn cannot promote.

$Fortress_{RP+B} = 1$ if P^c is Rook Pawn $\wedge B^c$ is wrong color for promotion square $\wedge K^c$ blockades.

- **Material vs. Fortress Interaction:**

$$Scale_{Fortress}^c = Mat^c \times Fortress_{RP+B}$$

3 Game Type and Board State Features

These features characterize the global state of the game. Unlike piece-specific features, these act as "context switches" for the linear model, allowing it to apply different weights to features depending on the phase (e.g., King safety is vital in the middlegame but King activity is vital in the endgame).

3.1 Game Phase Indicators

We define binary flags to categorize the game stage based on total non-pawn material.

- **Middlegame (Model):** 1 feature. Active if significant material remains on the board.

$$Phase_{MG} = 1 \quad \text{if Total Non-Pawn Material} > Threshold_{EG}.$$

- **Endgame (Normal):** 1 feature. Active if material has dropped below the threshold, indicating the transition to the endgame phase.

$$Phase_{EG} = 1 \quad \text{if Total Non-Pawn Material} \leq Threshold_{EG}.$$

- **Endgame (OCB):** 1 feature. Specific flag for "Opposite Color Bishop" endgames, which have a high draw tendency even with material advantage.

$$\text{Phase}_{OCB} = 1 \quad \text{if } \text{Phase}_{EG} = 1 \wedge B_{Light}^c = 1 \wedge B_{Dark}^{\bar{c}} = 1 \wedge \text{No other minors.}$$

- **Queens on Board:** 1 feature. The presence of Queens fundamentally changes the tactical sharpness of the position.

$$\text{HasQueens} = 1 \quad \text{if } Q^c = 1 \wedge Q^{\bar{c}} = 1.$$

3.2 Board Openness

These features classify the pawn structure to determine the value of long-range pieces (Bishops/Rooks) vs. short-range pieces (Knights). We determine openness based on the number of pawns remaining and the status of the central files.

- **Open Position:** 1 feature. Characterized by few pawns and open central files.

$$\text{Board}_{Open} = 1 \quad \text{if } \text{Pawn Count} < 8 \vee (\text{FileOpen}_d = 1 \wedge \text{FileOpen}_e = 1).$$

- **Closed Position:** 1 feature. Characterized by locked pawn chains and blocked centers.

$$\text{Board}_{Closed} = 1 \quad \text{if } \text{Center is locked (Ram features active)} \vee \text{Pawn Count} > 12.$$

- **Semi-Open Position:** 1 feature. The intermediate state.

$$\text{Board}_{Semi} = 1 \quad \text{if } \neg \text{Board}_{Open} \wedge \neg \text{Board}_{Closed}.$$

3.3 Strategic Scenarios

Additional global flags that dictate the nature of the struggle.

- **Opposite Castling:** 1 feature. Indicates a sharp attacking game where players are castled on opposite wings (e.g., White Short, Black Long). This typically devalues positional features and increases the weight of "Pawn Storm" and "Attack Speed" features.

$$\text{Scenario}_{OppC} = 1 \quad \text{if } K_{Kingside}^c \wedge K_{Queenside}^{\bar{c}} \text{ (or vice versa).}$$

3.4 Endgame Specifics

These features capture specific geometric and theoretical properties critical to navigating the endgame phase.

- **King Cut-Off:** 2 features per player. Indicates if a friendly Rook is preventing the enemy King from participating in the action by controlling a rank or file.

$\text{Cutoff}_{Rank}^c = 1$ if R^c controls a rank separating $K^{\bar{c}}$ from the main pawn masses or promotion squares.

$\text{Cutoff}_{File}^c = 1$ if R^c controls a file separating $K^{\bar{c}}$.

- **Opposition:** 1 feature per player. Captures whether the active player's King has the "opposition," a fundamental concept in pawn endgames.

$\text{Opposition}^c = 1$ if $\text{Dist}(K^c, K^{\bar{c}})$ is an odd number of squares on a rank/file, and it is the opponent's turn.

- **Lucena / Philidor Patterns:** 2 features per player. Binary flags for exact, known tablebase-like theoretical draws or wins.

$\text{Philidor}^c = 1$ if defending R^c is on the 3rd rank (relative) blocking the enemy King.